PROJECT DOCUMENTATION

2.4 GHz Rectangular Microstrip Patch Antenna

Design and Electromagnetic Simulation Using GNU Octave and openEMS

Project	Antenna design and simulation
Target frequency	2.4 GHz
Software	GNU Octave and openEMS
Feeding method	Inset microstrip feed; 50 Ω reference

1. Abstract

This project documents the design and electromagnetic simulation of a rectangular microstrip patch antenna intended to operate near 2.4 GHz. The model is developed using GNU Octave with the openEMS electromagnetic solver. The design uses a dielectric substrate with relative permittivity 3.38 and thickness 1.524 mm, an inset microstrip feed, and a 50 Ω reference impedance. The workflow covers antenna geometry, material definition, FDTD mesh generation, excitation and boundary setup, and post-processing of S_{11}, input impedance, and far-field radiation. Final numerical performance values must be entered from the verified simulation run.

2. Project Objectives

- Design a rectangular microstrip patch antenna for a target frequency of 2.4 GHz.
- Build a three-dimensional electromagnetic model from the specified dimensions.
- Implement an inset microstrip feed and a 50 Ω lumped-port reference.
- Run an FDTD simulation using openEMS through GNU Octave.
- Analyze S_{11}, resonant frequency, input impedance, impedance matching, and far-field radiation.
- Develop practical skills in mesh generation, port placement, boundary conditions, and simulation troubleshooting.

3. Design Specifications

The following are the specified model parameters. They are design inputs, not measured simulation results.

Parameter	Specified value
Target operating frequency	2.4 GHz
Relative permittivity, εr	3.38
Substrate thickness	1.524 mm
Substrate dimensions	80 mm × 80 mm
Patch length (x direction)	33.56 mm
Patch width (y direction)	42.23 mm
Feed-line width	3.52 mm
Inset depth	10.3 mm
Inset gap	1.0 mm
Reference impedance	50 Ω
Loss tangent	0.001

4. Antenna Structure and Feeding

The antenna model consists of a rectangular conducting patch on a dielectric substrate and a conducting ground plane on the opposite side. The substrate and ground-plane footprint is 80 mm × 80 mm. The patch measures 33.56 mm along x and 42.23 mm along y.

An inset microstrip feed is used. The feed line enters the patch region and terminates at an inset position. Changing the inset position changes the input impedance and can help match the antenna to the 50 Ω reference. The specified feed-line width is 3.52 mm.

The model uses a 50 Ω lumped port. Correct port placement and electrical connection to the feed are essential; an incorrectly positioned or disconnected port can lead to misleading S-parameter and impedance results.

5. Design Theory

5.1 Resonance and effective length

A rectangular microstrip patch resonates approximately according to its effective electrical length and the effective dielectric constant. Fringing fields at the radiating edges make the effective length slightly greater than the physical patch length.

5.2 Common first-pass design equations

- Patch width: $W = [c / (2f_0)] \sqrt{[2 / (\varepsilon r + 1)]}$.
- Effective permittivity: $\varepsilon eff \approx (\varepsilon r + 1)/2 + (\varepsilon r - 1)/2 \times [1 + 12h/W]^{-1/2}$.
- Length extension: $\Delta L/h \approx 0.412 \times [(\varepsilon eff + 0.3)(W/h + 0.264)] / [(\varepsilon eff - 0.258)(W/h + 0.8)]$.
- Effective length: $Leff = c / [2f_0\sqrt{\varepsilon eff}]$.
- Physical patch length: $L \approx Leff - 2\Delta L$.

Here c is the speed of light, f_0 is the target frequency, εr is relative permittivity, h is substrate thickness, and W is patch width. These equations provide initial dimensions; electromagnetic simulation is used to assess and tune the response. The dimensions in Section 3 are the project values supplied for this report and have not been independently recalculated here.

6. Simulation Environment and Method

6.1 Software

- GNU Octave: runs the MATLAB-compatible script and performs post-processing.
- openEMS: performs the electromagnetic FDTD simulation.

6.2 Geometry and materials

The script defines the dielectric substrate, conducting patch, ground plane, and inset feed geometry. The dielectric is assigned the specified relative permittivity and loss tangent; the patch and ground are represented as conducting regions.

6.3 FDTD configuration

The project script configuration described in the development notes uses a maximum of 80,000 time steps, an end-criteria value of 1×10^{-5}, and PML_8 absorbing boundaries. Confirm the final script for the exact excitation bandwidth, mesh settings, and port geometry before treating this report as a complete record of a particular run.

6.4 Frequency sweep and post-processing

The post-processing configuration evaluates the response from 2 GHz to 3 GHz using 1,001 frequency points. Port wave quantities are used to calculate S_{11}, and port voltage/current quantities are used to calculate input impedance. Far-field post-processing is used to inspect radiation characteristics.

7. Antenna Performance Measures

7.1 Reflection coefficient and return loss

S_{11} describes the reflected wave relative to the incident wave at the input port. Its magnitude in decibels is $20 \log_{10}|S_{11}|$. A more negative value indicates less reflected signal. The −10 dB level is commonly used as a reference when determining impedance bandwidth.

7.2 Resonant frequency

For this project, identify the resonant frequency from the frequency at which $|S_{11}|$ reaches its minimum in the final verified sweep. Do not assume that the simulated resonance is exactly 2.4 GHz because 2.4 GHz is the design target.

7.3 Input impedance

Input impedance is Z_in = R + jX, where R is resistance and X is reactance. Near a well-matched resonance for a 50 Ω reference, the impedance is expected to be close to 50 + j0 Ω. Record the actual value from the final simulation.

7.4 Radiation pattern and directivity

The far-field radiation pattern shows how radiation varies with direction. Directivity describes radiation concentration relative to an isotropic radiator. Verify whether the software output is linear or in dBi before reporting it. Do not call directivity gain unless antenna efficiency is included in the calculation.

8. Simulation Results — Complete After Final Run

Use values and plots from one clean, final simulation run. Do not combine results from different runs or use illustrative/example values.

Quantity	Final verified result
Resonant frequency	[Enter measured value in GHz]
Minimum S_{11}	[Enter measured value in dB]
Input impedance at resonance	[Enter R + jX in Ω]
−10 dB bandwidth	[Enter measured range or bandwidth]
Directivity	[Enter verified value and units, if available]
Gain	[Enter only if calculated and verified]

Figures to include

- Antenna geometry screenshot showing patch, substrate, ground plane, and inset feed.
- S_{11} / return-loss plot with a −10 dB reference line.
- Input-impedance plot showing resistance and reactance.
- 2D far-field radiation pattern.
- 3D far-field radiation pattern, if generated and verified.

9. Discussion

The antenna response depends on the patch dimensions, dielectric properties, feed location, port definition, mesh resolution, and absorbing boundary conditions. Mesh quality is especially important near metal edges, substrate interfaces, and the feed region, where fields can change rapidly. If the simulated resonance differs from the target, review the geometry, material values, feed, mesh, and boundary setup before changing dimensions.

A low S_{11} value alone does not prove that every antenna requirement is satisfied. Input impedance and radiation characteristics should also be checked, and the results should be interpreted together.

10. Conclusion

This project establishes a GNU Octave/openEMS workflow for modeling and simulating a 2.4 GHz rectangular microstrip patch antenna with an inset microstrip feed. It combines initial antenna dimensioning, electromagnetic geometry construction, FDTD setup, port analysis, and far-field evaluation. The work provides practical experience with antenna modeling and simulation troubleshooting. Final claims about resonance, matching, bandwidth, directivity, and gain should be added only after verification against the final simulation output.

11. Suggested GitHub Repository Contents

- README.md — project summary, specifications, software requirements, and run instructions.
- Microstrip_antenna_design.m — final commented GNU Octave/openEMS script.
- results/ — actual plots and geometry screenshots from the final run.
- documentation/ — this report and supporting theoretical calculations.

12. Reproducibility Checklist

- Run the script from a clean simulation directory and confirm it completes successfully.
- Remove machine-specific paths or explain how to update them.
- Record the GNU Octave and openEMS versions used.
- Ensure all plots and numerical results come from the same final run.
- Verify units and definitions for S_{11}, impedance, directivity, and gain.
- Do not publish placeholder or illustrative values as measured simulation results.

Scope note: This report is based on the design specifications and workflow details available in the project discussion. It is a structured project document, not a substitute for the final source script or raw simulation data. Complete the marked result fields using the actual final run.